Supplemental Material: Reliability-aware active learning for FFLO phase spaces

Author list pending human approval

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PHYSICAL MODEL AND FREE-ENERGY FUNCTIONAL

We study a one-dimensional altermagnetic superconductor in the Nambu basis $\Psi_k = (c_{k+q/2,\uparrow}, c_{k+q/2,\downarrow}, c^\dagger_{-k+q/2,\uparrow}, c^\dagger_{-k+q/2,\downarrow})^T$ with

$$H_{\mathrm{BdG}}(k) = \begin{pmatrix} h(k+q/2) - \mu & \Delta_q \\ \Delta_q^* & -\sigma_y h^T(-k+q/2)\sigma_y + \mu \end{pmatrix}, \tag{1}$$

where $h(k) = t\cos k + (0, \lambda_y \sin k, \lambda_z \sin k + J_A \cos k) \cdot \boldsymbol{\sigma}$. The implementation evaluates the normal-referenced discrete free energy

$$F(\Delta, q) = \frac{1}{2N_k}\sum_{k,n} E_n(k;\Delta,q) f[E_n/(k_B T)] + \frac{\Delta^2}{U} - \frac{1}{2N_k}\sum_{k,n} E_n(k;0,q) f[E_n(k;0,q)/(k_B T)], \tag{2}$$

with $f(z) = 1/(e^z + 1)$ in a numerically stable sigmoid form. The final term defines the normal reference. All energies are reported in units of t.

EXACT BDG ORACLE

Algorithm S1: Exact thermodynamic and topology oracle. For each requested coordinate: (1) construct the BdG spectrum on the coarse (Δ, q) grid; (2) form the normal-referenced free energy; (3) detect global, near-degenerate, and edge-risk minima; (4) expand the q window directionally when the low-energy structure is not enclosed, up to three expansions; (5) refine selected local basins using at least 800 q points and 600 Δ points in the adaptive-box path; (6) compare near-zero-Δ states with a strictly positive-Δ scan; (7) assign the thermodynamic phase and a separate numerical status; (8) return flagged points for retention or rerun rather than silently treating them as reliable; and (9) if the state is reliable, gapped, and FFLO, evaluate the full-zone gap and Pfaffians. Duplicate coordinates are reconciled by the configured four-decimal key before append.

TABLE S1. Physical and numerical parameters.

Quantity	Value	Role
t	1	energy unit
λ_y, λ_z	0.6, 0.6	spin–orbit couplings
μ	0.55	2D chemical potential
U	1.6	pairing interaction
ω_D	10	momentum mask cutoff
N_k	800	Brillouin-zone grid
coarse Δ grid	$[0, 0.6]$, 400 points	thermodynamic scan
initial q grid	$[-1, 0.5]$, 400 points	momentum scan
$\Delta_\epsilon, q_\epsilon$	$10^{-3}, 10^{-2}$	phase assignment

RELIABILITY-STATE HIERARCHY

Accepted/reliable points have an enclosed minimum and a resolved normal-state comparison. Boundary-delicate points lie within the adopted finite energy or order-parameter resolution. A q-window-unresolved point retains low-energy structure at the scanned edge after allowed expansion; a near-zero-Δ-unresolved point lacks a resolved condensation-energy sign. Rerun-required records remain outside the clean physical training view. Topology-inapplicable includes all non-FFLO or numerically unreliable states, while gap-unresolved states do not receive a $\mathbb{Z}_2$ label. Phase identity, training eligibility, rerun status, and topology applicability are separate fields.

COMPLETE ACTIVE-LEARNING LOOP

Algorithm S2: Reliability-aware active learning. (1) Draw a 512-point random-grid cold start from the full candidate domain. (2) Construct the training-eligible exact view and a phase-stratified 85/15 training/validation split. (3) Train independent regression and classification ensembles from scratch with deterministic member seeds. (4) predict all unseen candidate locations; (5) compute acquisition components and apply the candidate mask, exact-coordinate exclusion, active-pool rule, and observation repulsion; (6) construct a

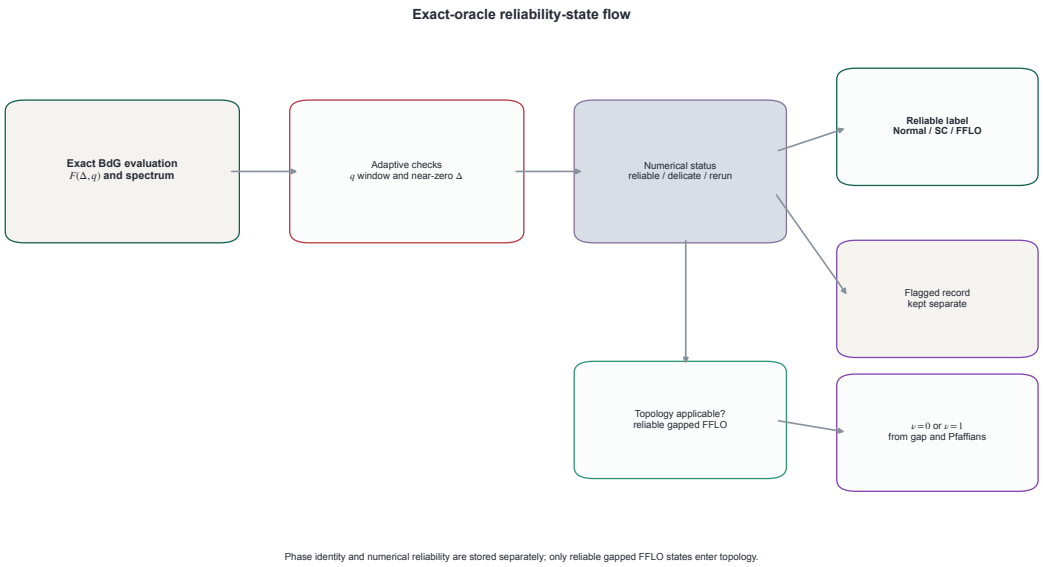


FIG. S1. Exact-oracle reliability-state flow. Numerical status is separate from phase identity, and topology is restricted to reliable gapped FFLO states.

stochastic microbatch with within-batch repulsion; (7) write restartable point shards; (8) run the exact BdG oracle; (9) merge returned shards and separate reliable, delicate, and rerun records; (10) append eligible unique observations; and (11) recompute fixed-grid diagnostics and continue until an operational stop or budget limit. Checkpoints include the dataset, selection tables, run configuration, monitor predictions, stop history, and shard metadata. Stage III adds topology applicability and spectral acquisition components; Stage IV replaces the Cartesian 2D pool by a three-coordinate candidate construction but retains exact-label authority.

SURROGATE ARCHITECTURE AND TRAINING

The implementation uses separate PyTorch ensembles for regression and phase classification, not a Gaussian process or tree model. Each of five members is an MLP with two 64-unit hidden layers and ReLU activations. Inputs are standardized by the training mean and

standard deviation. Regression targets $(\Delta, q, \eta, I_c^+, I_c^-)$ are also standardized; the regression loss is mean-squared error. The classifier uses cross entropy on Normal, uniform-SC, and FFLO labels. Adam uses learning rate 10^{-3} and weight decay 10^{-5}, batch size 512, and 240 epochs for both tasks. Members use seeds $42 + 1000 + i$ and $42 + 2000 + i$ for regression and classification. Models are retrained from scratch at each scheduling step; no early-stopping rule is implemented in these model classes. Regression uncertainty is the member standard deviation; phase probabilities are averaged over members and converted to entropy and margin scores.

TABLE S2. Surrogate and training hyperparameters.

Item	Value
ensemble size	5 per task
hidden layers	2×64, ReLU
validation fraction	0.15, phase stratified
optimizer	Adam
learning rate / weight decay	$10^{-3}/10^{-5}$
epochs / batch size	240 / 512
regression loss	MSE
classification loss	cross entropy
uncertainty	member dispersion / averaged probabilities

ACQUISITION FUNCTION AND MICROBATCH SELECTION

Let $U_{\mathrm{cls}} = 0.4H + 0.6\exp[-m/0.2]$, where H is normalized phase entropy and m is the largest-probability margin. Let $U_{\mathrm{reg}} = (U_\Delta + U_q)/2$, $B_{NS} = \exp[-|\hat{\Delta} - \Delta_\epsilon|/0.04]\, 4P_N P_{SC}$, $B_{UF} = P_{SC}\exp[-||\hat{q}| - q_\epsilon|/0.04]$, and $G = (G_\Delta + G_q)/2$. The production profile is

$$A_{\mathrm{phase}} = U_{\mathrm{cls}} + 0.6U_{\mathrm{reg}} + B_{NS} + 0.9B_{UF} + 0.5G, \qquad A_0 = A_{\mathrm{phase}} + 0.4E_q + w_{\mathrm{ext}}E_{\mathrm{ext}}, \quad (3)$$

where E_q flags predicted q-edge risk and E_{ext} combines extrapolation with uncertainty. w_{ext} decreases piecewise from 0.15 to 0.08 and 0.03 at iterations 10 and 30. High-confidence interiors receive a soft penalty; the active pool is set by the larger of a scheduled quantile and

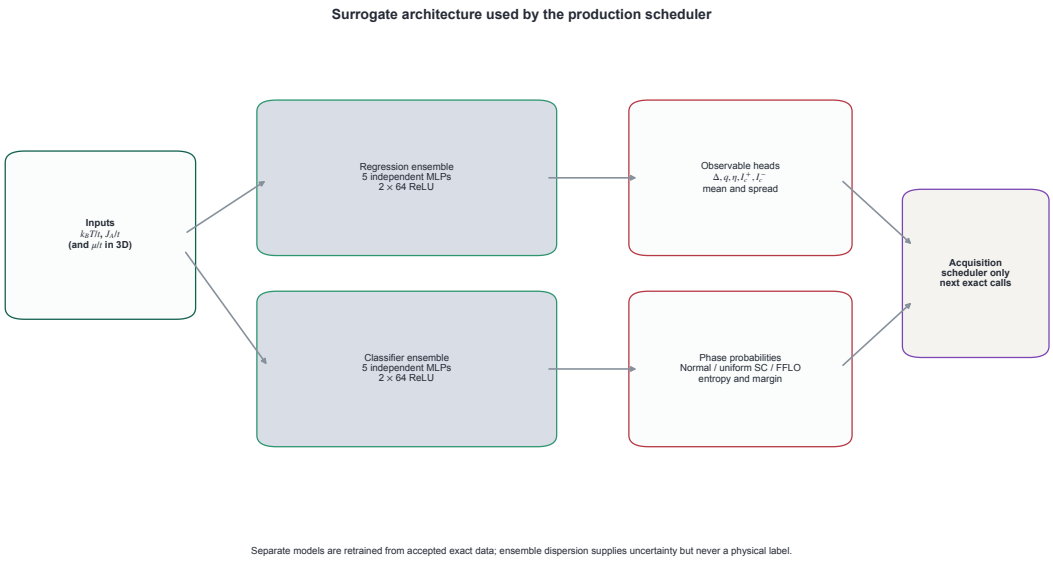


FIG. S2. Production surrogate architecture. The regression and classification ensembles supply scheduling scores, not physical labels.

$0.7p_{95}(A_0)$. Observation repulsion uses length 0.02 and floor 0.5. During greedy stochastic construction, batch repulsion uses length 0.03 and floor 0.01, yielding $A_{\text{select}} = A_0 R_{\text{obs}} R_{\text{batch}}$ and sampling probability proportional to $A_{\text{select}}^{\gamma}$, with $\gamma = 1.5, 2.5, 4.0$ across the same schedule. Existing exact coordinates are excluded before selection.

Algorithm S3: Repelled microbatch construction. Build the active pool from A_0 after masks and interior penalties; multiply by observation repulsion; sample one coordinate using the scheduled power; update R_{batch} around that coordinate; repeat until the requested batch is reached or the high-information pool is exhausted.

SAMPLING EFFICIENCY

The Stage-I candidate universe is the complete 241×321 Cartesian grid over $k_BT/t \in [0, 0.56]$ and $J_A/t \in [0, 2.12]$, containing 77,361 locations. The final primary discovery dataset contains 6,880 unique exact coordinates. Thus $f_{\text{exact}} = 6880/77361 = 0.08893$

TABLE S3. Production acquisition components.

Component	Definition	Normalization or weight	Role
phase entropy/margin	U_{cls}	0.4/0.6 mixture	classification ambiguity
regression spread	U_{reg}	min–max normalized; 0.6	phase-facing uncertainty
Normal–SC	B_{NS}	competition gated; weight 1	order-parameter boundary
uniform–FFLO	B_{UF}	SC gated; weight 0.9	finite-q boundary
gradients	G	grid derivative; weight 0.5	rapid variation
numerical risk	E_q	weight 0.4	q-window risk
exploration	E_{ext}	annealed 0.15/0.08/0.03	domain coverage
repulsion	$R_{\mathrm{obs}}R_{\mathrm{batch}}$	length/floor rules	avoid redundant calls

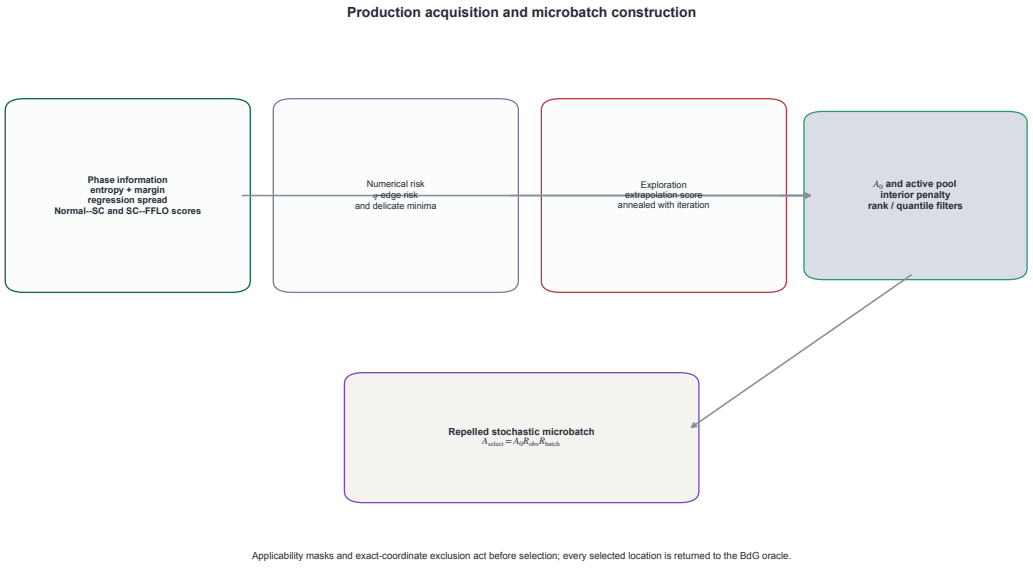


FIG. S3. Acquisition decomposition and repelled microbatch selection.

and $R_{\mathrm{call}} = 77361/6880 = 11.24$. The archived selection tables contain 8,192 call records and 7,305 unique selected coordinates; these counts differ because returned, accepted, and duplicate-accounting layers are not interchangeable. A boundary-enrichment factor cannot

be reconstructed because the export does not retain complete per-iteration candidate-pool boundary labels. Per-iteration wall times exist, but a matched exhaustive-grid baseline does not, so no wall-time speedup is claimed.

TABLE S4. Sampling and exact-call accounting.

Quantity	Value
candidate locations	77,361
selected call records	8,192
unique selected coordinates	7,305
unique exact coordinates	6,880
training-eligible records	6,880
reliable exact records	6,104
sampling fraction	8.893%
exact-location reduction	11.244×
boundary enrichment	not recoverable
wall-time speedup	not established

NUMERICAL DIAGNOSTICS AND OPERATIONAL STOPPING

These criteria are operational controls for the numerical workflow. They are not thermodynamic observables or universal physical scales. Phase-map change is the fraction of fixed monitor-grid cells whose predicted phase changes between iterations. Boundary displacement is a bidirectional nearest-neighbor p_{95} distance between successive extracted boundary point sets after normalization by the parameter-domain ranges. Coverage is the p_{95} distance from the current monitor-grid boundaries to exact samples. Surprise is the fraction of returned labels that differ from the prior prediction, with trusted and flagged denominators stored separately. Four of five main conditions must hold for four successive iterations after a five-iteration minimum. The adopted thresholds reflect the monitor-grid resolution and workflow tolerance, not a physical scale.

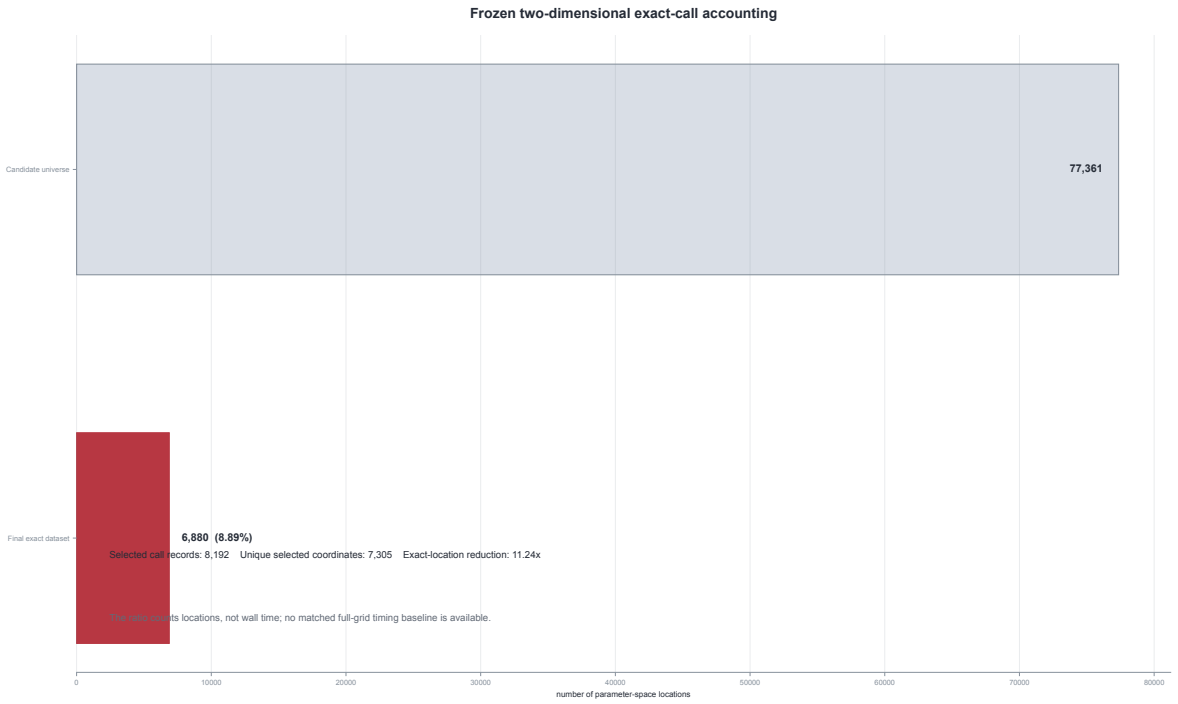


FIG. S4. Frozen candidate-universe and exact-location accounting. The reduction is a location-count ratio, not a wall-time acceleration.

TABLE S5. Operational diagnostics.

Diagnostic	Definition	Domain	Operational setting	Interpretation
phase-map change	changed-cell fraction	fixed 241×321 monitor	< 0.002	map stability
Normal–SC shift	bidirectional p_{95}	normalized domain	< 0.004167	boundary stability
SC–FFLO shift	bidirectional p_{95}	normalized domain	< 0.004167	boundary stability
coverage	boundary-to-exact p_{95}	normalized domain	< 0.00625	sampling support
label surprise	prediction/return mismatch	selected calls	< 0.05	sequential stability
patience	successive eligible iterations	iteration history	4	operational termination

TOPOLOGY DIAGNOSIS AND RECONSTRUCTION SENSITIVITY

Topology is applicable only to thermodynamically accepted, numerically reliable, gapped FFLO states. The minimum absolute BdG eigenvalue is scanned over the full Brillouin zone.

At $k = 0, \pi$, the analytic Pfaffians are

$$P_0 = (\mu - t\cos(q/2))^2 + \Delta^2 - (\lambda_y \sin(q/2))^2 - (J_A \cos(q/2) + \lambda_z \sin(q/2))^2, \tag{4}$$

$$P_\pi = (\mu + t\cos(q/2))^2 + \Delta^2 - (\lambda_y \sin(q/2))^2 - (J_A \cos(q/2) + \lambda_z \sin(q/2))^2. \tag{5}$$

Away from the relative Pfaffian tolerance 10^{-8}, $P_0 P_\pi < 0$ gives $\nu = 1$ and $P_0 P_\pi > 0$ gives $\nu = 0$. Analytic values were checked against direct Majorana-basis Pfaffians, and CPU/GPU full-zone gaps were compared in the archived topology validation. The displayed transition contour is a deterministic support-restricted reconstruction on a 401×401 monitor grid with support radius 0.045. For neighborhoods $k = 6, 8, 12$, its arc lengths are 0.8270, 0.8215, and 0.8038, while coverage p_{95} is 0.006269, 0.006185, and 0.005913. This sensitivity and the local opposite-$\mathbb{Z}_2$ brackets make the contour a guide to the gap-closing region rather than an exact boundary; nodal alternatives are not exhaustively excluded.

THREE-DIMENSIONAL EXTENSION

Stage IV samples $(k_B T/t, J_A/t, \mu/t)$ with $\mu/t \in [-0.5, 1.5]$. The main figure shows the exact thermodynamic point cloud, reliable FFLO topology support, and bins centered at $\mu/t = -0.45, 0.15, 0.85$, and 1.45 with half-width 0.025. No interpolation is performed between slices. Archived map-change, proxy-surface displacement, coverage, surprise, and component-count histories are retained as numerical diagnostics only. The discrete support reveals systematic three-dimensional structure but does not establish a converged continuous topology surface or complete atlas.

REPRODUCIBILITY, CODE, DATA, AND ENVIRONMENT

The source registry maps exact evaluation, surrogate training, acquisition, stopping, and topology to public modules. Frozen configuration: Python 3.10.20, PyTorch 2.11.0+cu130, seed 42, and eight H100 shards with round-robin partitioning. Archived runs retain configurations, selected coordinates, exact shards, dataset views, checkpoints, monitor predictions, and stop history. Regeneration uses `figures/build_publication_supplement_figures.py` and `manuscript_langgraph/scripts/run_public_manuscript_reproducible.py`. Because the export lacks Git metadata, SHA-256 hashes identify inputs and outputs.